



**TRIBHUVAN UNIVERSITY  
INSTITUTE OF ENGINEERING  
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**A  
MID TERM REPORT  
ON**

**“ LSTM-BASED APPROACH FOR GENERATOR STATE  
PREDICTION IN TRANSIENT STABILITY ASSESSMENT AND  
CONTROL USING TRANSIENT ENERGY FUNCTION “**

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# TABLE OF CONTENTS

|                                                                |           |
|----------------------------------------------------------------|-----------|
| <b>1.INTRODUCTION</b>                                          | <b>1</b>  |
| <b>2.LITERATURE REVIEW</b>                                     | <b>3</b>  |
| 2.1. Traditional Transient Stability Assessment Methods        | 3         |
| 2.2. The Transient Energy Function                             | 3         |
| 2.3. Hybrid Framework for Real-Time Control (The Base Paper)   | 6         |
| 2.4. Machine Learning for Enhanced Time Series Prediction      | 7         |
| <b>3.PROBLEM STATEMENT, OBJECTIVES, SCOPE &amp; LIMITATION</b> | <b>9</b>  |
| 3.1. Problem Statement                                         | 9         |
| 3.2. Project Objectives                                        | 9         |
| 3.3. Scope of the Project                                      | 9         |
| 3.4. Limitations                                               | 10        |
| <b>4.METHODOLOGY AND RELATED THEORY</b>                        | <b>11</b> |
| 4.1. Offline Computation/Look-up Table :                       | 11        |
| 4.2.Online Computation                                         | 12        |
| 4.3.Algorithm to calculate $t_{cr}$ and $V_{cr}$               | 14        |
| <b>5.SYSTEM UNDER CONSIDERATION, SCOPE &amp; SOFTWARE</b>      | <b>15</b> |
| 5.1. System Under Consideration                                | 15        |
| 5.2.Scope                                                      | 16        |
| 5.3.Software                                                   | 17        |
| <b>6.RESULT AND ANALYSIS</b>                                   | <b>18</b> |
| 6.1.Transient Energy Function and BCU Results From MATLAB      | 18        |
| 6.2.Time Domain Simulation Using PSS/E                         | 21        |
| <b>7.REMAINING WORK</b>                                        | <b>23</b> |
| <b>8.CONCLUSION</b>                                            | <b>24</b> |
| <b>9.REFERENCES</b>                                            | <b>25</b> |

## ***LIST OF FIGURES***

|                                                                                   |           |           |
|-----------------------------------------------------------------------------------|-----------|-----------|
| <i>Fig 1: Region of attraction and computation of tcr.</i>                        | <i>4</i>  |           |
| <i>Fig 2: Potential Energy “well”</i>                                             | <i>6</i>  |           |
| <i>Fig 3: PEBS</i>                                                                | <i>6</i>  |           |
| <i>Fig 4: Block Diagram of Offline Computation</i>                                | <i>11</i> |           |
| <i>Fig 5: Block Diagram of Online Computation</i>                                 | <i>12</i> |           |
| <i>Fig 6: Flowchart for generation shedding</i>                                   | <i>13</i> |           |
| <i>Fig 7: IEEE 9 Bus Network</i>                                                  | <i>15</i> |           |
| <i>Fig 8: IEEE-39 Bus Network</i>                                                 | <i>16</i> |           |
| <i>Fig 9: w vs del for faulted and Post fault Dynamics (Stable)</i>               |           | <i>18</i> |
| <i>Fig 10: del vs t and w vs t for faulted and Post fault Dynamics (Stable)</i>   |           | <i>19</i> |
| <i>Fig 11: w vs del for faulted and Post fault Dynamics (Unstable)</i>            |           | <i>20</i> |
| <i>Fig 12: del vs t and w vs t for faulted and Post fault Dynamics (Unstable)</i> |           | <i>20</i> |
| <i>Fig 13: Plot 1[time vs rotor angle]</i>                                        |           | <i>21</i> |
| <i>Fig 14: Plot 2[Time vs rotor angle]</i>                                        |           | <i>21</i> |
| <i>Fig 15: plot 3 [time vs speed]</i>                                             |           | <i>22</i> |
| <i>Fig 16: Plot 4[time vs speed]</i>                                              |           | <i>22</i> |

# ABSTRACT

Transient stability remains a critical challenge in modern power systems due to the increasing complexity of grid operations and the presence of fast, large disturbances. The Transient Energy Function (TEF), combined with the Boundary of Controlling Unstable Equilibrium Point (BCU) method, offers a fast and reliable alternative to conventional time-domain simulation for stability assessment. However, real-time emergency control actions such as generation shedding are limited by computational and communication delays, requiring accurate prediction of generator rotor angles and speeds at the control actuation instant. The existing curve-fitting approach used for this prediction suffers from poor accuracy during highly non-linear first-swing dynamics.

This project enhances the TEF-based real-time control framework by integrating a Long Short-Term Memory (LSTM) neural network for state prediction. LSTMs are capable of learning complex, non-linear, time-dependent behaviors and thus provide superior approximation of generator dynamics during the post-fault period. The methodology includes offline computation of  $V_{cr}$  and  $t_{cr}$  using the BCU method, development of normalized kinetic-energy signatures for MOD identification, and online assessment using real-time PMU data. Preliminary results on the IEEE 9-bus and IEEE 39-bus systems demonstrate accurate stability assessment and agreement between TDS, PEBS, and BCU evaluations. The proposed LSTM-enhanced prediction module is expected to significantly improve the reliability and speed of emergency control decisions, paving the way for ultra-fast transient stability management in large interconnected power systems.

# 1.INTRODUCTION

Power system Stability has always been a great concern from its developmental phase. Power System stability refers to the system's ability to return to normal or stable operation after having been subjected to some form of disturbance. In practical operations, power systems inevitably experience various disturbances such as short-circuit faults, line disconnections, and the integration or disconnection of large-capacity loads. These disturbances frequently lead to widespread blackouts globally. Power system Stability has been classified into various types as Voltage Stability, Frequency Stability, Resonance Stability and Angle Stability (Transient Stability).

Our project is related to Angle Stability during large disturbance i.e Transient Stability. Transient stability involves the response to large disturbances, which may cause rather large changes in rotor speeds, power angles, and power transfers. It is a fast phenomenon usually evident within a few seconds.

Transient stability assessment methodologies are broadly classified into three main types:

- **Simulation Methods** (Time-Domain Simulation)
- **Direct Methods** (Energy Function based)
- **Data-Driven Methods** (Machine Learning based)

Our work focuses on the **Direct Method**, which leverages the **Transient Energy Function (TEF)** integrated with computational techniques to determine system stability quickly and accurately. The TEF framework provides a robust metric for assessing the system's total energy (kinetic and potential). To assess stability, this calculated energy must be compared against a **critical energy limit** ( $V_{cr}$ ), which represents the maximum energy the system can absorb without losing synchronism. This critical threshold is accurately determined offline using the Boundary of Controlling Unstable Equilibrium Point (BCU) methodology.

This project is built upon the real-time stability prediction and control framework proposed by Pratyasa Bhui and Nilanjan Senroy [1]. Their hybrid method combines an **offline** database of stability boundaries (computed using the BCU technique) with an **online** assessment driven by PMU data. Following the detection of an unstable disturbance, the system determines the required emergency control action, typically generation shedding.

A critical step in this control phase is the need to predict the generator states (rotor angle and speed) at a future time-specifically, the moment the control signal (e.g., generation shedding) is expected to take effect. This prediction compensates for the inherent computational and communication delay ( $T_d$ ) in the control loop. The base methodology achieves this prediction using **cubic and quadratic curve fitting** over a small window of post-fault data.

However, the non-linear nature of power system swings especially during the first, critical swing limits the predictive accuracy of simple polynomials. To enhance the reliability and precision of the emergency control action, **we propose replacing the conventional curve fitting technique with a Long Short-Term Memory (LSTM) neural network**. LSTMs are inherently capable of learning complex, time-dependent, and non-linear patterns within time-series data. Once trained on a comprehensive dataset of fault trajectories, the LSTM model can provide significantly more accurate prediction of generator angle and speed at the point of control actuation, thereby improving the estimation of the required generation shedding and minimizing the risk of unnecessary or insufficient control action.

This enhancement aims to deliver an ultra-fast, accurate, and reliable framework for transient stability assessment and emergency control on complex systems, demonstrated using the IEEE 39-Bus test system.

## 2.LITERATURE REVIEW

Transient stability refers to the power system's ability to remain in synchronism following a large, sudden disturbance, such as a severe fault or the loss of a major transmission line. The assessment of transient stability is critical for preventing widespread cascading failures and blackouts, necessitating methodologies that are both accurate and rapid enough to facilitate emergency control actions.

### 2.1. Traditional Transient Stability Assessment Methods

Historically, power system stability has been analyzed using two main approaches:

#### 2.1.1. Time-Domain Simulation (TDS)

TDS involves the numerical integration of the system's differential and algebraic equations over time, providing the most detailed and accurate representation of rotor angle and speed trajectories [5]. However, the computational complexity of TDS renders it too slow for real-time applications. For large-scale power grids, the time required to complete a full simulation often exceeds the critical time window available for making and executing emergency control decisions.

#### 2.1.2. Direct Methods and the Transient Energy Function (TEF)

To overcome the speed limitation of TDS, Direct Methods were developed. These methods rely on the Transient Energy Function (TEF), a single scalar function that describes the system's total energy (kinetic and potential) for a post fault system[4].

### 2.2. The Transient Energy Function

Power system undergoing disturbance can be modeled by set of three differential equation as given in [5] as:

$$\begin{aligned}\dot{x}(t) &= f^I(x(t)) & -\infty < t \leq 0 \\ \dot{x}(t) &= f^F(x(t)) & 0 < t \leq t_{cl} \\ \dot{x}(t) &= f(x(t)) & t_{cl} < t < \infty\end{aligned}$$

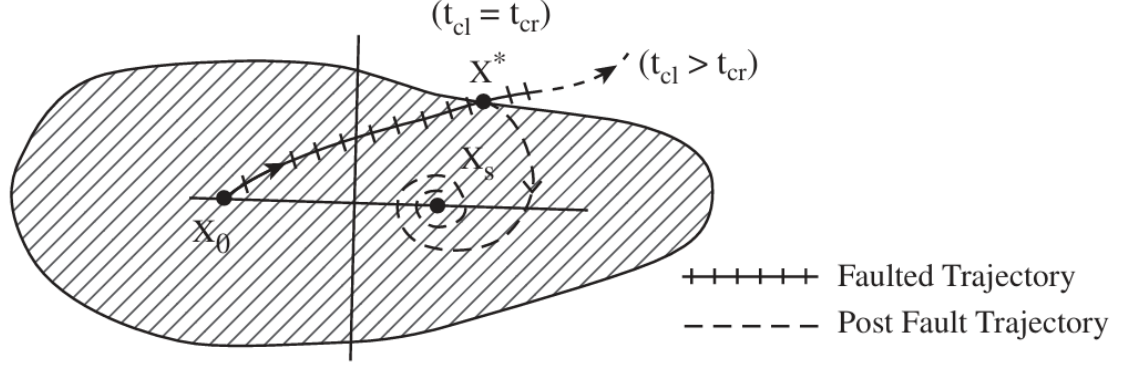


Fig 1: Region of attraction and computation of  $t_{cr}$ .

The swing equation for multi-machine system is given by [5]:

$$\frac{2H_i}{\omega_s} \frac{d^2 \delta_i}{dt^2} + D_i \frac{d\delta_i}{dt} = P_{mi} - P_{ei}, \quad i = 1, \dots, m$$

where

$$P_{ei} = E_i^2 G_{ii} + \sum_{\substack{j=1 \\ j \neq i}}^m (C_{ij} \sin \delta_{ij} + D_{ij} \cos \delta_{ij}).$$

Energy function is given by [5]:

$$V(\delta, \omega) = \frac{1}{2} M \omega^2 + V_{PE}(\delta).$$

The energy function of multimachine system is given by [5]:

$$\begin{aligned} V(\theta, \tilde{\omega}) &= \frac{1}{2} \sum_{i=1}^m M_i \tilde{\omega}_i^2 - \sum_{i=1}^m \int_{\theta_i^s}^{\theta_i} f_i(\theta) d\theta_i \\ &= \frac{1}{2} \sum_{i=1}^m M_i \tilde{\omega}_i^2 - \sum_{i=1}^m P_i(\theta_i - \theta_i^s) - \sum_{i=1}^{m-1} \sum_{j=i+1}^m \left[ C_{ij}(\cos \theta_{ij} - \cos \theta_{ij}^s) \right. \\ &\quad \left. - \int_{\theta_i^s + \theta_j^s}^{\theta_i + \theta_j} D_{ij} \cos \theta_{ij} d(\theta_i + \theta_j) \right] \end{aligned}$$

Transient Energy is the total energy in the power system after the fault is cleared. It is the sum of potential and kinetic energy in the system for a post fault system.



The Transient energy function for the post fault system is described by following function[1]:

$$V(\omega_i, \theta_i) = V_{KE} + V_{PE} = \{0.5 M_{eq} \omega_{eq}^2\} - \left\{ \sum_{i=1}^m P_i (\theta_i - \theta_i^s) + \sum_{i=1}^{m-1} \sum_{j=i+1}^m [C_{ij} (\cos \theta_{ij} - \cos \theta_{ij}^s) + D_{ij} \frac{\theta_i - \theta_i^s + \theta_j - \theta_j^s}{\theta_{ij} - \theta_{ij}^s} (\sin \theta_{ij} - \sin \theta_{ij}^s)] \right\}$$

where,  $\omega_i$ =speed of  $i^{th}$  generator,  $M_{eq}$ = moment of inertia of two machine equivalent,  $\omega_{eq}$ = speed of two machine equivalent,  $E$  =

internal emf,  $P_i = P_{mi} - E_i^2 G_{ii}$ ,  $\theta^s$ =post fault stable equilibrium point, m=number of generators,  $C_{ij} = E_i E_j B_{ij}$ ,  $D_{ij} = E_i E_j G_{ij}$ .

The energy function consists of four terms-

- $0.5 M_{eq} \omega_{eq}^2$ : total corrected kinetic energy of the generators.
- $\sum_{i=1}^m P_i (\theta_i - \theta_i^s)$ : total potential energy of the generators.
- $\sum_{i=1}^{m-1} \sum_{j=i+1}^m C_{ij} (\cos \theta_{ij} - \cos \theta_{ij}^s)$ : stored magnetic in all branches
- $\sum_{i=1}^{m-1} \sum_{j=i+1}^m D_{ij} \frac{\theta_i - \theta_i^s + \theta_j - \theta_j^s}{\theta_{ij} - \theta_{ij}^s} (\sin \theta_{ij} - \sin \theta_{ij}^s)$ : approximate dissipated energy of all branches.

The stability determination within the TEF method is based on a straightforward comparison: the system is stable if the energy gained during the fault,  $V_{cl}$  is less than the critical energy limit,  $V_{cr}$ . i.e.  $V_{cl} < V_{cr}$

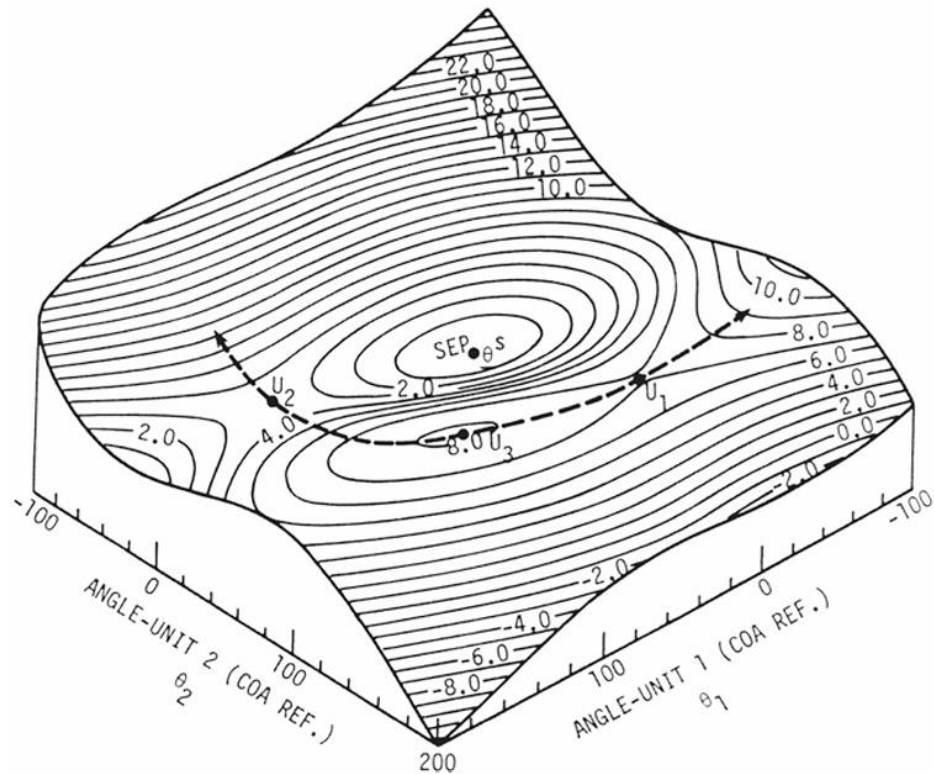
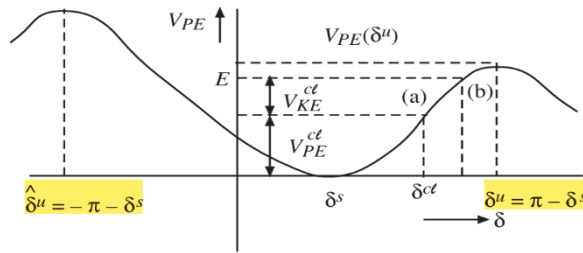
The swing equation can be modeled as below for further analysis[1]:

$$M_i \frac{d^2 \theta_i}{dt^2} = P_i - \sum_{j=1, \neq i}^n (C_{ij} \sin \theta_{ij} + D_{ij} \cos \theta_{ij}) - \frac{M_i}{M_T} P_{coi} = f_i(\theta), \quad i = 1, \dots, m$$

where,  $P_i = P_{mi} - E_i^2 G_{ii}$

$$P_{coi} = \sum_{i=1}^m P_i - 2 \sum_{i=1}^m \sum_{j=i+1}^m D_{ij} \cos \theta_{ij}$$

- Potential Energy Boundary Surface (PEBS): An initial, faster approximation method that estimates the critical energy by tracking the potential energy along the fault trajectory. While quick, PEBS often provides an inaccurate estimate of the true stability limit.



- **Boundary of Controlling Unstable Equilibrium Point (BCU):**The BCU methodology is the high-accuracy technique used to rigorously locate the Controlling Unstable Equilibrium Point (CUEP). The potential energy at the CUEP defines the true Vcr for a specific fault [4]. The BCU method is essential for obtaining a reliable stability threshold.

### 2.3. Hybrid Framework for Real-Time Control (The Base Paper)

Our project is founded on the real-time prediction and control framework proposed by Bhui and Senroy [1]. This hybrid methodology effectively addresses the computational hurdles of the Direct Method to enable quick, real-time stability assessment:

### 2.3.1. BCU-TEF Hybrid Structure

The framework operates in two modes:

1. **Offline Computation:** The highly accurate BCU method is used offline to compute  $V_{cr}$  and critical clearing time  $T_{cr}$  for various fault locations and system topologies. This helps to find the correct MOD (Mode of Disturbance). Corresponding to each MOD Normalized kinetic energy Vector is calculated which acts as signature for online Computation. This data is stored in a look-up table.
2. **Online Assessment:** When an actual fault occurs, real-time data from Phasor Measurement Units (PMUs) are used to quickly identify the corresponding MOD from the look-up table and determine the stability margin ( $V_{cr} - V_{cl}$ ) with high confidence followed by proper control action [1].

### 2.3.2. The Curve Fitting Bottleneck

The effectiveness of the framework relies on the ability to take immediate emergency control action (e.g., generation shedding) if the system is found to be unstable. However, an inherent computational and communication delay ( $T_d$ ) exists between the moment instability is detected and the moment the control action is physically executed.

To compensate for this delay, the base paper requires the prediction of the generator's state (rotor angle and speed) at the future control actuation time,  $T_{cl} + T_d$ . The original method employs cubic and quadratic curve fitting over a small post-fault time window to extrapolate the generator states.

This reliance on polynomial curve fitting constitutes the primary bottleneck in the system's predictive accuracy. The non-linear, oscillatory dynamics of the power system's first swing are poorly approximated by simple polynomial functions, leading to prediction errors in angle and speed. Since kinetic energy ( $V_{ke}$ ) is proportional to the square of speed  $w^2$ , small errors in the speed prediction are amplified, compromising the accuracy of the final generation shedding calculation.

## 2.4. Machine Learning for Enhanced Time Series Prediction

To resolve the predictive limitations of polynomial fitting, this project leverages advancements in deep learning for time series data.

### 2.4.1. Long Short-Term Memory (LSTM) Networks

Long Short-Term Memory (LSTM) networks are a specialized architecture of Recurrent Neural Networks (RNNs) designed specifically to model complex, sequential data by overcoming the vanishing gradient problem [3]. Using internal memory cells and gating mechanisms (input, forget, and output gates), LSTMs excel at learning long-term dependencies and hidden non-linear relationships within a time series. The internal structure of the LSTM can be given as below.

### **2.4.2. Justification for LSTM Integration**

Given the need to accurately forecast the oscillatory and transient behavior of generator angles and speeds, the LSTM model offers a theoretically superior solution to curve fitting. Once trained on a comprehensive dataset of fault trajectories, the LSTM can model the non-linear swing dynamics with high fidelity. This leads to a more precise prediction of the generator state at the control actuation time, directly translating into a more accurate estimation of the required generation shedding amount. LSTMs are already successfully employed in the power domain for other time-series tasks, such as load and renewable generation forecasting [2].

By integrating the LSTM prediction engine into the control phase of the TEF framework, we aim to significantly improve the reliability and precision of emergency stability control in modern power systems.

## 3.PROBLEM STATEMENT, OBJECTIVES, SCOPE & LIMITATION

### 3.1. Problem Statement

The primary challenge is to develop a highly accurate, real-time-capable state prediction module that minimizes forecast error during the critical control delay period, ensuring optimal sizing of the emergency control action.

### 3.2. Project Objectives

The primary objectives of this project are derived from the need to enhance the predictive accuracy of the control module within the established TEF framework:

1. To develop and implement an LSTM (Long Short-Term Memory) neural network model for accurate, real-time prediction of generator angles and speeds.
2. To integrate the generator state predictions into the Transient Energy Function (TEF) framework for improved transient stability assessment and control
3. To demonstrate the applicability of the enhanced TEF method on the IEEE 39-bus test system

### 3.3. Scope of the Project

1. **System Model:** The project focuses on the transient stability analysis of the IEEE 39-bus New England Test System.
2. **Core Focus:** The project's novel contribution is strictly confined to the enhancement of the prediction module (replacing the curve fitting method with LSTM) within the existing control calculation framework.
3. **Control Action:** The emergency control action under consideration is selective generation shedding applied to the most unstable generators.

### 3.4. Limitations

The following inherent limitations are acknowledged in the execution and applicability of this project:

1. **Classical Model Assumption:** Generator Model for the TEF is a classical model for simplicity.
2. **Network Model:** Internal node model of the network is used in the project.
3. **Data Dependency:** The LSTM model is a data-driven approach. Its superior performance is critically dependent on the quality, quantity, and comprehensiveness of the training data (i.e., the simulated fault trajectories). The model may perform poorly on fault types or system conditions not represented in the training set.
4. **Real-Time PMU Data:** The online assessment assumes the availability of synchronized, high-speed, high-quality data from Phasor Measurement Units (PMUs) to feed the TEF and LSTM.

## 4.METHODOLOGY AND RELATED THEORY

### 4.1. Offline Computation/Look-up Table :

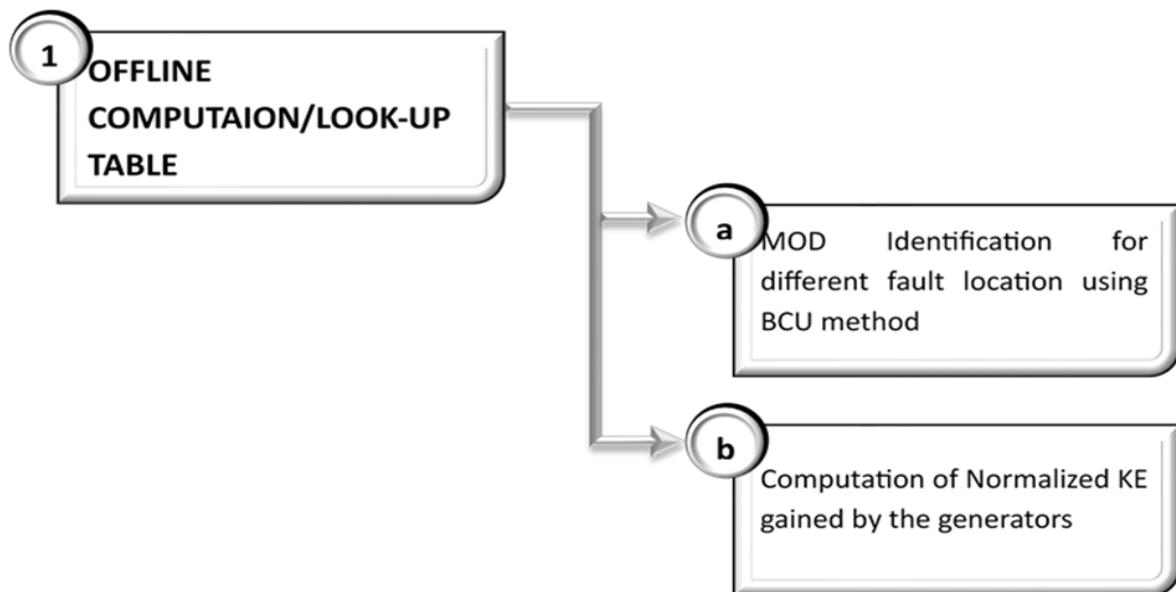


Fig 4: Block Diagram of Offline Computation

#### a. MOD Identification for different fault location

MOD is obtained using BCU Method:

- i. Tcr is calculated using BCU method.
- ii. Time domain simulation is done for a fault duration slightly higher than Tcr.
- iii. Detection of peak of the potential energy and generator ranking descending order of the angles.
- iv. Taking top machines as MOD and calculate CCT using MOD technique.
- v. machine groups with minimum error between time domain simulating CCT and TEF CCT taken as the MOD.

#### b. Computation of normalized KE gained by the generators:

Kinetic energy Vector is given by:

$$KE_X = [ke_{1x} \ ke_{2x} \ ... \ ke_{ix} \ .. \ ke_{mx}]$$

Normalized Kinetic Energy Vector is given by:

$$\begin{aligned} \overline{KE}_X &= [\overline{ke}_{1x} \ \overline{ke}_{2x} \ .. \ \overline{ke}_{ix} \ .. \ \overline{ke}_{mx}] \\ &= \frac{[ke_{1x} \ ke_{2x} \ ... \ ke_{ix} \ .. \ ke_{mx}]}{\sum_{i=1}^m ke_{ix}} \end{aligned}$$

## 4.2. Online Computation

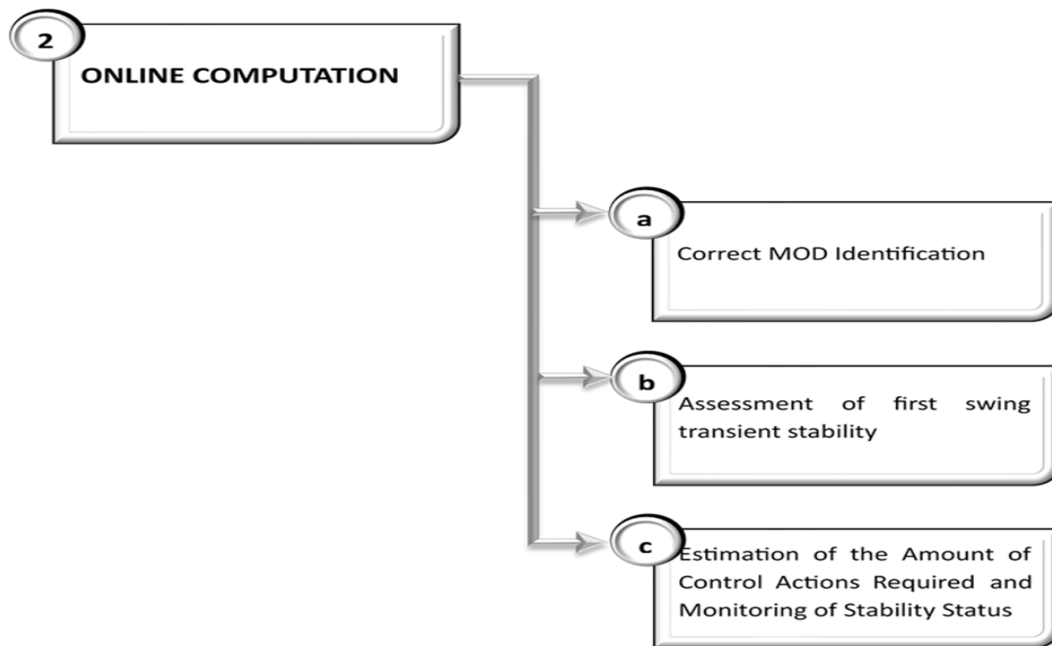


Fig 5: Block Diagram of Online Computation

### a. Correct Mode Identification

**Step 1 :** Calculation of KE

**Step 2 :** Ranking of Probable MODs

**Step 3 :** Calculation of Post-Fault SEP

**Step 4 :** Calculation of CUEP

**Step 5 :** Identification of correct MOD

### b. Assessment of first swing transient stability

$$\Delta V_{cl} = V_{cr} - V_{cl} \quad \begin{array}{l} >0, \text{ Stable} \\ <0, \text{ Ustable} \end{array}$$

### c. Estimation of Amount of Control Actions Required and Monitoring of Stability Status

**Step 1 :** Prediction of generator speed and angles[LSTM]

**Step 2 :** Modification in Energy Function for Generation Shedding

**Step 3 :** Estimation of the amount of Control action required



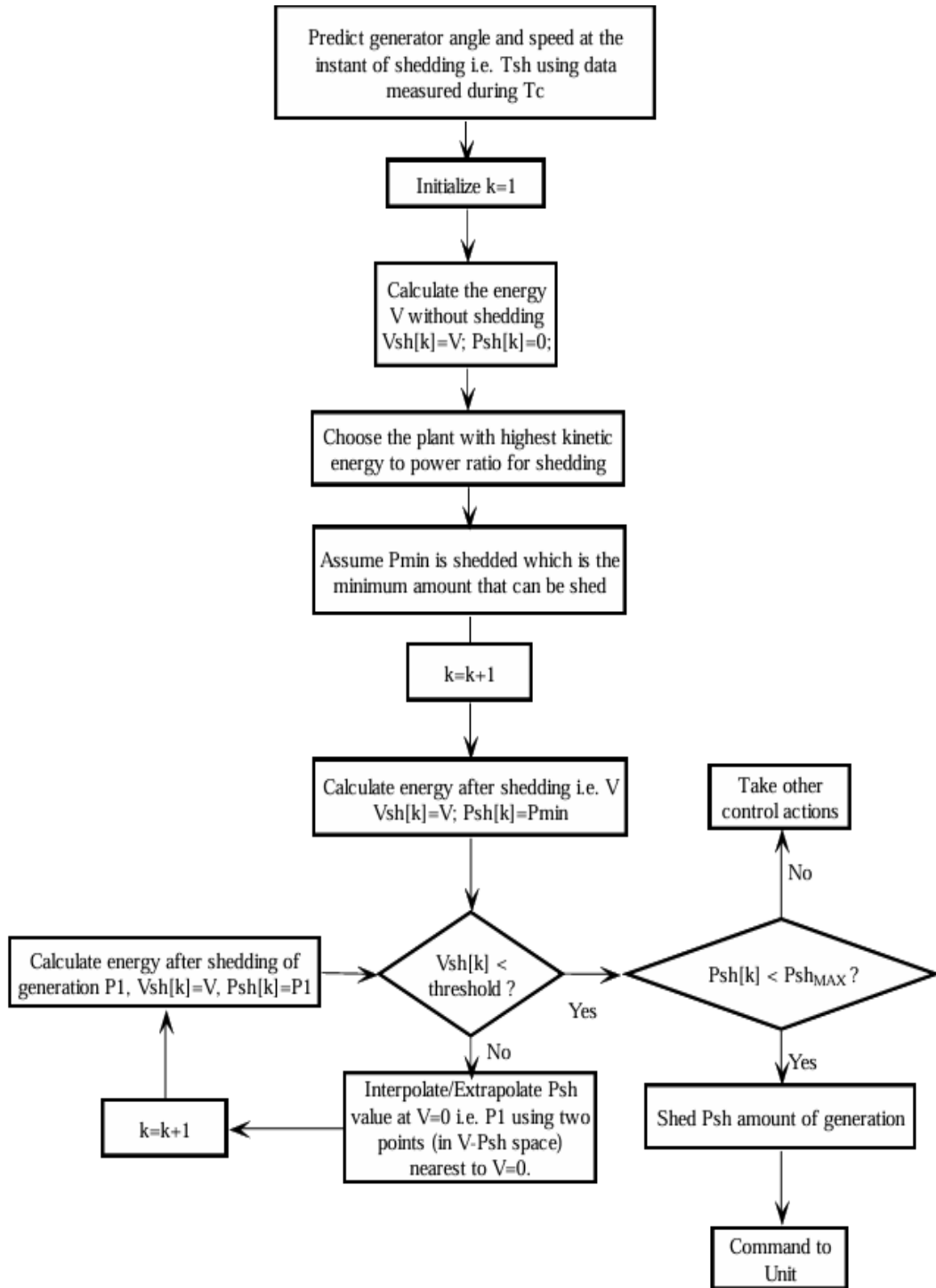


Fig 6: Flowchart for generation shedding

### 4.3. Algorithm to calculate tcr and Vcr

- Preprocessing and network reduction
- Build pre-fault, post-fault network
- Kron reduction to generate network only
- Compute  $B_{ij}$ ,  $G_{ij}$ ,  $E'$  and  $P$  terms
- Non-linear swing Dynamical Model
- Direct Time Domain Simulation (TDS)
- Energy Function (TEF)
- PEBS Method:
  - a. Integrate fault on dynamic until tcr
  - b. Compute PEBS function
  - c. PEBS Crossing
  - d. Compute Critical Energy (PEBS)
  - e. Critical clearing time estimate
- BCU (Unstable Equilibrium Search):
  - a. Initialize PEBS point and applying artificial dynamics
  - b. Identify UEP
  - c. Compute Critical Energy (BCU)
  - d. Critical clearing time
- Final Decision (Energy margin)

## 5.SYSTEM UNDER CONSIDERATION, SCOPE, AND SOFTWARE

### 5.1. System Under Consideration

#### IEEE 9 BUS System

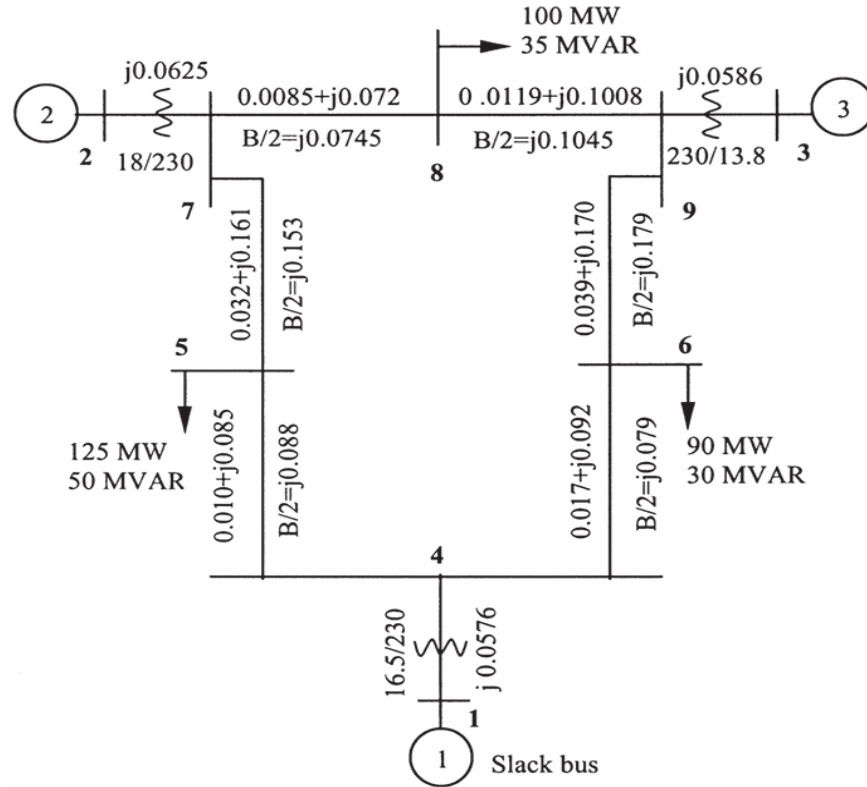


Fig 7: IEEE 9 Bus Network

The IEEE 9-bus test system is a compact benchmark model widely used for studying transient stability, power flow, and control techniques. This system represents a simple, stable and balanced three-machine network. It consists of:

- 9 buses
- 3 synchronous generators (at buses 1, 2, and 3)
- 3 generator step-up transformers
- 6 transmission lines
- 3 load buses

## IEEE39 Bus System

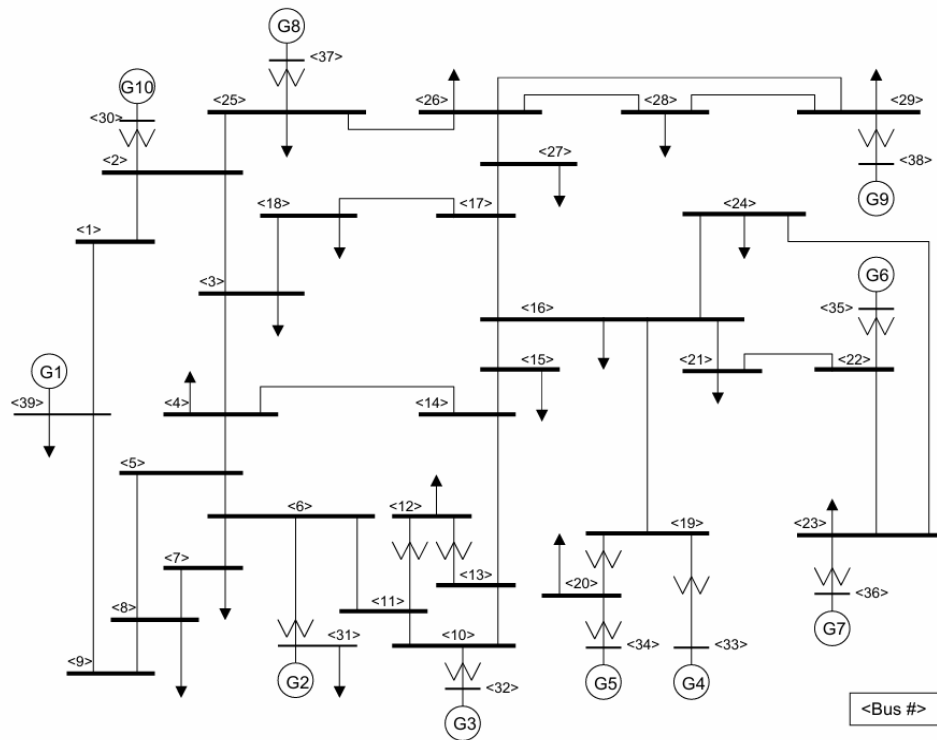


Fig 8: IEEE-39 Bus Network

The IEEE-39 bus system, also known as the New England 10-machine system, represents a medium-to-large-scale interconnected power grid. It is characterized by:

- 39 Buses
- 10 Synchronous Generators (G1 to G10)
- 12 Transformers and 46 Transmission Lines

## 5.2.Scope

### Network Modeling:

For the development of the TEF Internal node model is developed using the constant impedance load and internal node behind the transient reactance is used followed by Kron's reduction to show the coupling between the generators. For the development of the real system all the parameters are included.

**Generator Modeling:** Classical Model in TEF and Detailed Model in PSS/E

**Disturbance Type:** 3-Phase Symmetrical Fault, Bus Fault

### **5.3.Software**

#### **Matlab:**

MATLAB plays a crucial role in transient stability analysis by providing a flexible platform to model power system components, simulate disturbances, and solve the swing equations that describe generator dynamics. Using tools such as Simulink and Simscape Electrical, engineers can create detailed system models, apply faults, and observe the system's response in real time. MATLAB's numerical solvers enable accurate computation of rotor angle and speed deviations, while its visualization capabilities help evaluate system stability and determine critical clearing times. Overall, MATLAB offers an efficient and comprehensive environment for analyzing and improving the transient stability of electrical power systems.

#### **Python:**

Python is increasingly used in transient stability analysis due to its flexibility, open-source ecosystem, and powerful scientific libraries. With packages like NumPy, SciPy, and Pandas, Python efficiently solves the swing equations and performs numerical integration for multi-machine systems. Visualization tools such as Matplotlib enable clear plotting of rotor angle, speed, and power variations following disturbances, while libraries like PyPSA, GridCal, and Pandapower provide advanced power-system modeling and fault-simulation capabilities. Python also supports automation, batch simulations, optimization, and machine-learning-based stability assessment. Overall, Python offers an accessible, customizable, and cost-effective environment for comprehensive transient stability studies.

#### **PSS/E:**

PSS®E (Power System Simulator for Engineering) is a widely used industry-standard software for performing transient stability analysis and other advanced power system studies. It provides highly detailed and accurate modeling of generators, networks, controls, and protection systems, allowing engineers to simulate faults, switching events, and dynamic system behavior with precision. PSS/E includes robust numerical solvers for swing equations, extensive model libraries, and powerful scripting capabilities through Python for automation. Its reliability, speed, and comprehensive analysis tools make it a preferred platform for utilities, researchers, and grid operators in evaluating system stability and planning secure operation.

## 6.RESULT AND ANALYSIS

### 6.1.Transient Energy Function and BCU Results From MATLAB

Transient energy function framework is developed for IEEE-9 bus system using classical model consideration. Fault at different location is created and results obtained is shown in the table below:

#### Result 1

| Fault At bus 7 at t=0 and cleared at 0.15 |                  |                       |                        |
|-------------------------------------------|------------------|-----------------------|------------------------|
| Result                                    | TDS              | BCU                   | PEBS                   |
| V                                         | $V_{cl} = 0.843$ | $V_{cr\_BCU} = 0.973$ | $V_{cr\_PEBS} = 1.25$  |
| Critical clearing time (Tcl)              | $T_{cl} = 0.15$  | $T_{cr\_BCU} = 0.162$ | $T_{cr\_PEBS} = 0.184$ |
| Computation Time                          | 0.23             | 0.2                   | 0.17                   |

$$V_{cr\_PEBS} > V_{cl}$$

$T_{cr\_PEBS} > T_{cl}$       System is PEBS Stable

$$V_{cr\_BCU} > V_{cl}$$

$T_{cr\_BCU} > T_{cl}$       System is BCU Stable

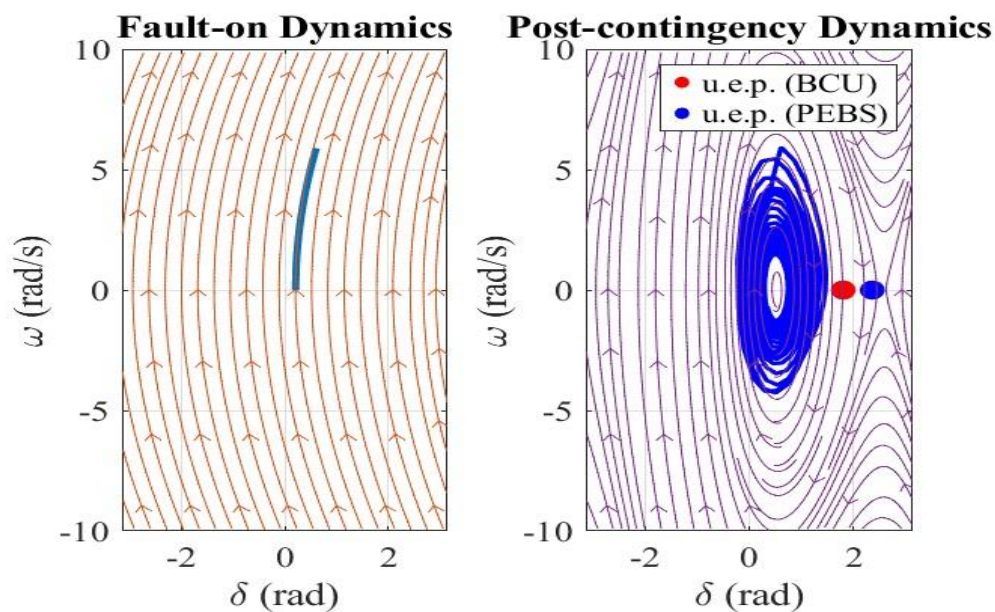


Fig 9:  $w$  vs  $\delta$  for faulted and Post fault Dynamics (Stable)

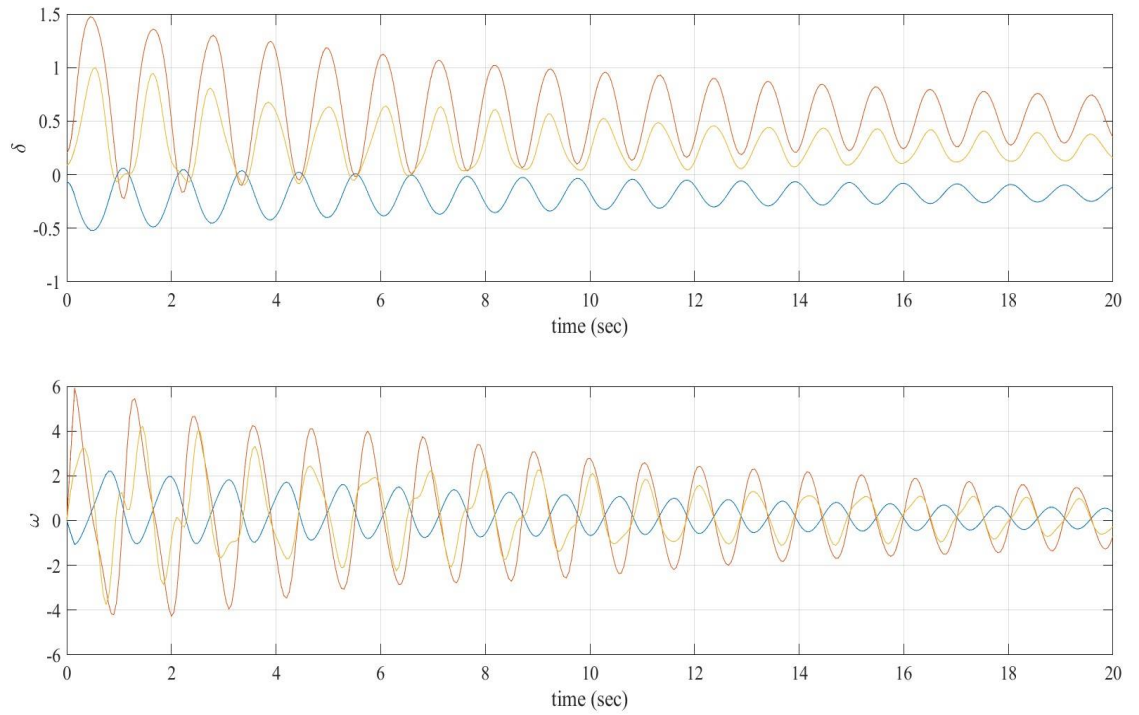


Fig 10:  $\delta$  vs  $t$  and  $w$  vs  $t$  for faulted and Post fault Dynamics (Stable)

## Result 2

| Fault At bus 7 at $t=0$ and cleared at 0.2 |                |                     |                      |
|--------------------------------------------|----------------|---------------------|----------------------|
| Result                                     | TDS ()         | BCU                 | PEBS                 |
| V critical                                 | $V_{cl}=1.473$ | $V_{cr\_BCU}=0.973$ | $V_{cr\_PEBS}=1.25$  |
| Critical clearing time (Tcl)               | $T_{cl}=0.2$   | $T_{cr\_BCU}=0.162$ | $T_{cr\_PEBS}=0.184$ |
| Computation Time                           | 0.15           | 0.1                 | 0.07                 |

$$V_{cr\_PEBS} > V_{cl}$$

$T_{cr\_PEBS} > T_{cl}$       System is PEBS Unstable

$$V_{cr\_BCU} > V_{cl}$$

$T_{cr\_BCU} > T_{cl}$       System is BCU Unstable

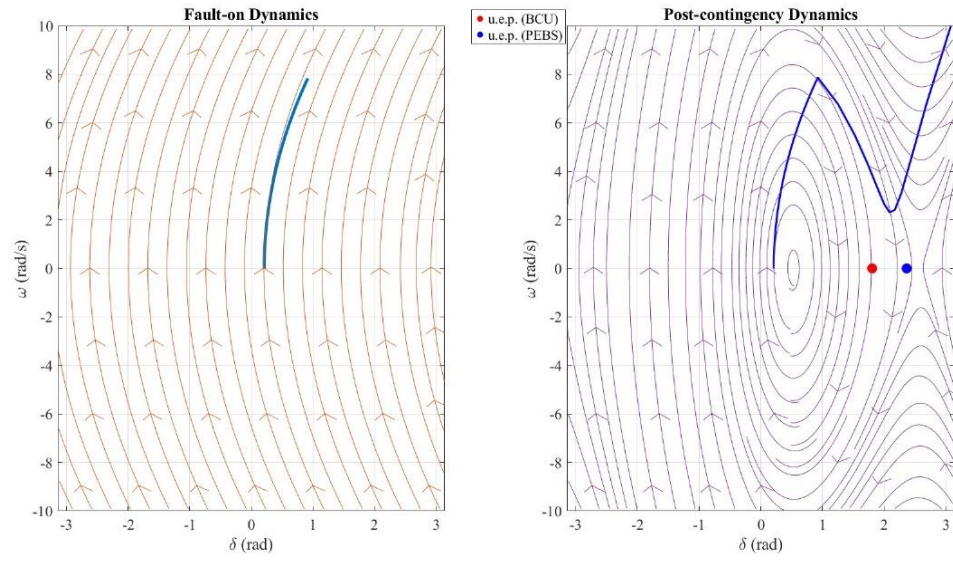


Fig 11:  $w$  vs  $\delta$  for faulted and Post fault Dynamics (Unstable)

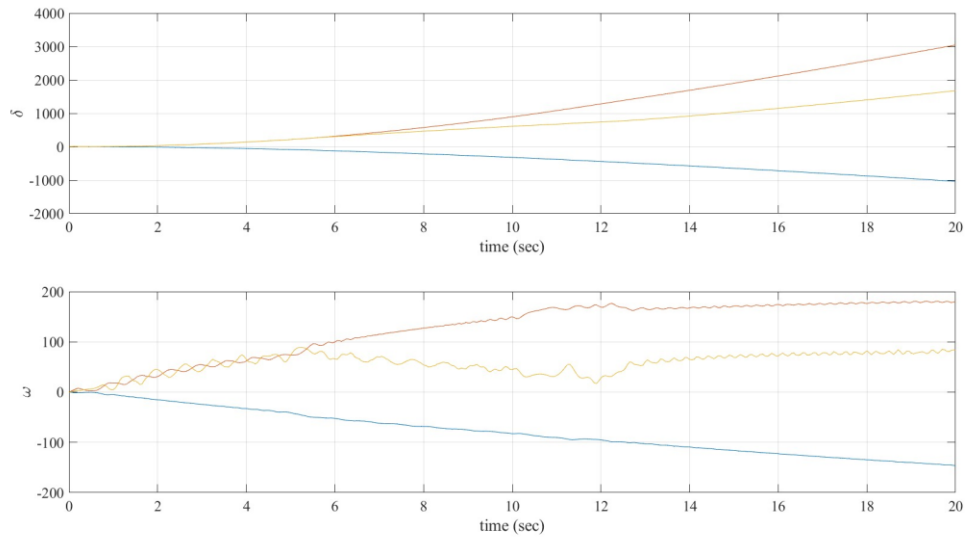


Fig 12:  $\delta$  vs  $t$  and  $w$  vs  $t$  for faulted and Post fault Dynamics (Unstable)



## 6.2. Time Domain Simulation Using PSS/E

### Output 1:

Fault created at Bus 29 at  $t=1.0$  and cleared at  $t=1.4$

Generator[G9] goes out of synchronism

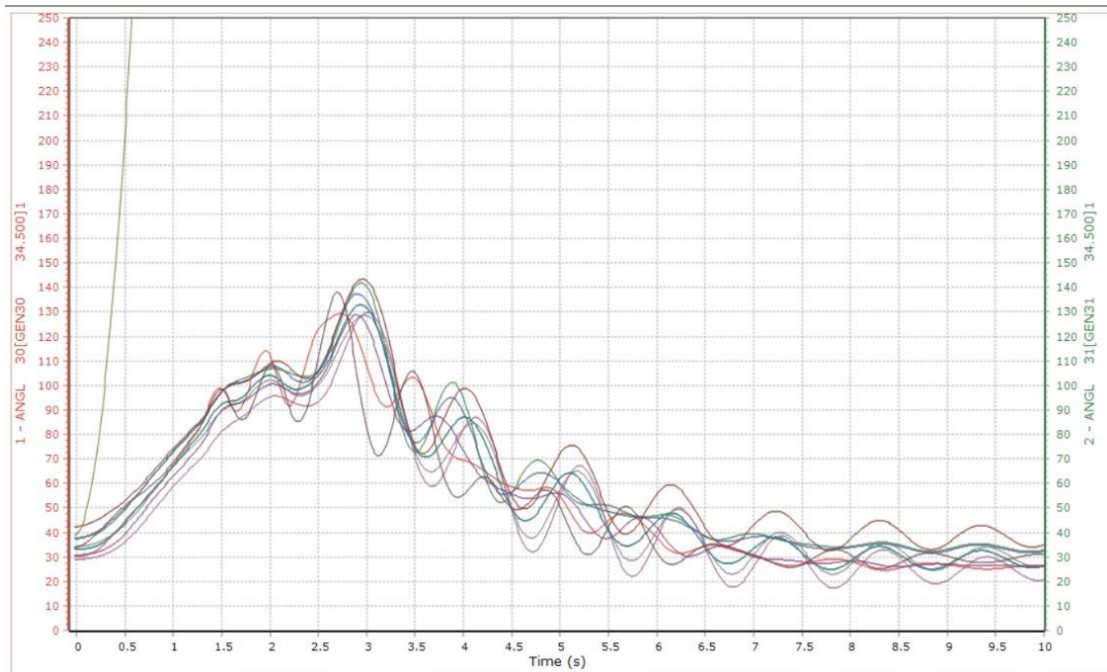


Fig 13: Plot 1[time vs rotor angle]

### Output 2:

Fault created at bus 2 at  $t=1.0$  and cleared at  $t=1.4$ . G10 goes out of synchronism.

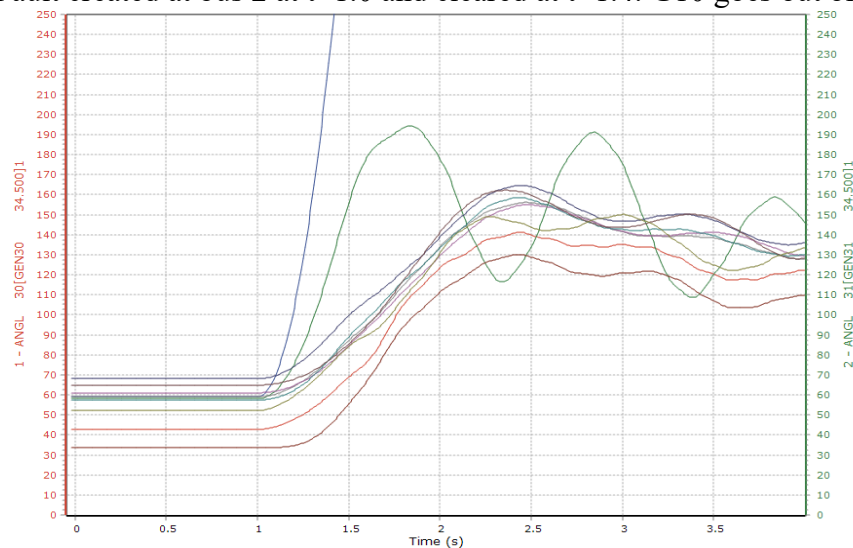


Fig 14: Plot 2[Time vs rotor angle]

Output 3:

Fault created at Bus 29 at  $t=1.0$  sec and cleared at  $t=1.3$

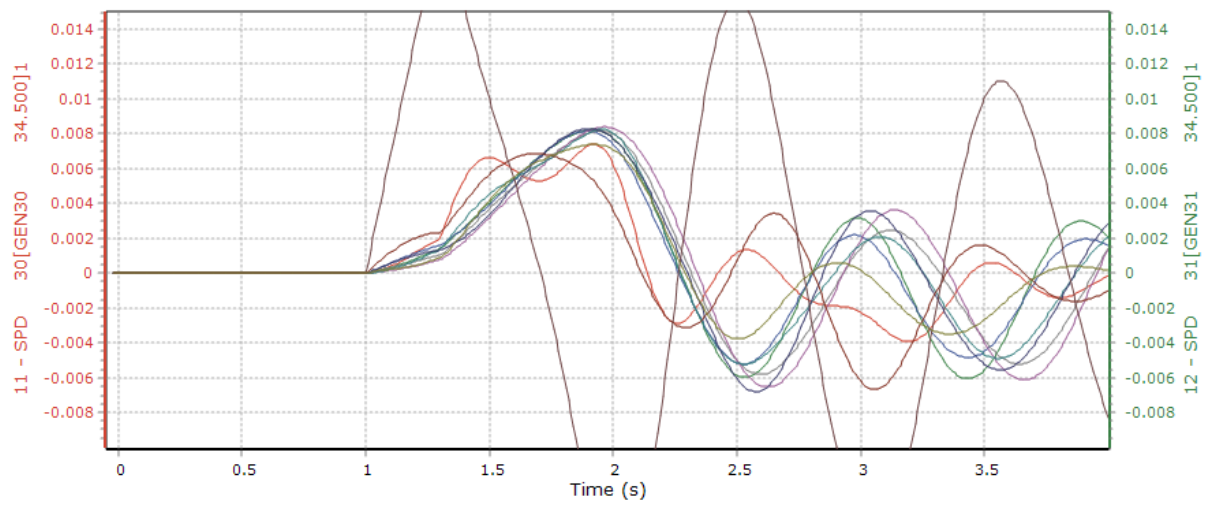


Fig 15: plot 3 [time vs speed]

Output 4:

Fault created at bus 10 at  $t=1.0$  sec and cleared at  $t=1.3$  sec.

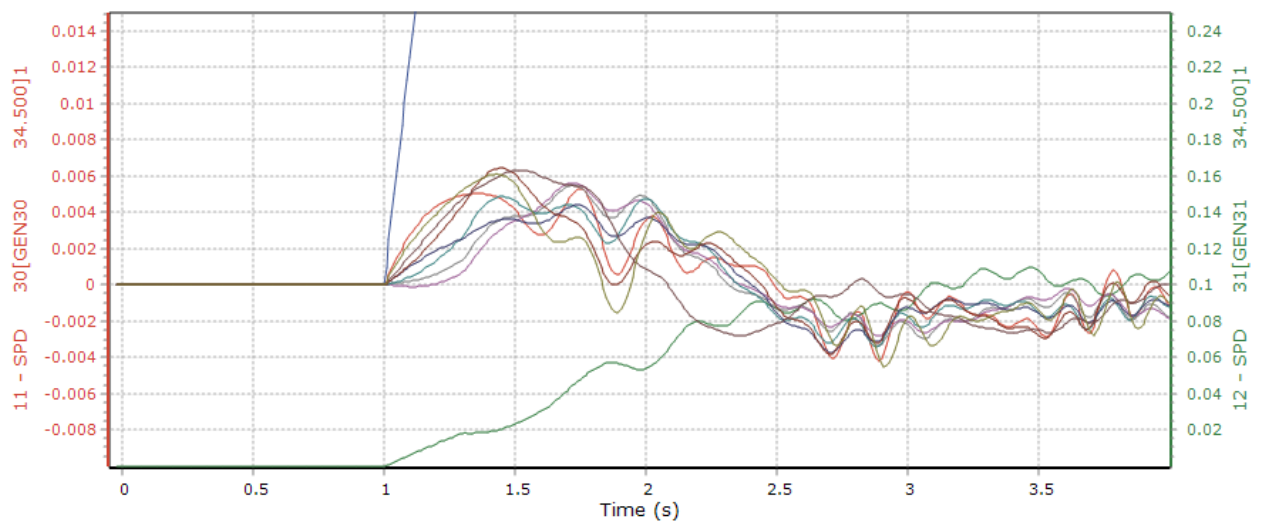


Fig 16: Plot 4[time vs speed]

## **7.REMAINING WORK**

- Lookup Table
- LSTM
- Model Training
- Integration of LSTM with TEF for control action

## **8.CONCLUSION**

Although the current stage has focused on developing the TEF-BCU framework and validating its behavior on benchmark systems, the remaining tasks construction of the lookup table, LSTM model training, and full integration of the prediction-control pipeline will complete the proposed enhanced stability assessment system. Upon completion, this method holds strong potential for deployment in real-time grid monitoring systems that rely on PMU data, contributing to a more resilient, faster-reacting, and smarter power grid.

## 9. REFERENCES

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